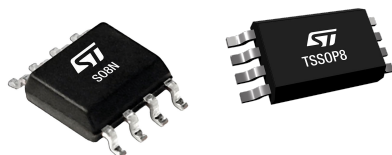


512-Kbit serial I²C bus EEPROM with configurable device address, software write protection registers and preprogrammed device address

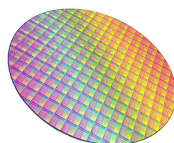


SO8N
150 mil width

TSSOP8
169 mil width



UFDFPN8
DFN8 - 2x3 mm



Unseen wafer

Features

I²C interface

- Compatible with the following I²C bus modes:
 - 1 MHz (Fast-mode Plus)
 - 400 kHz (Fast-mode)
 - 100 kHz (Standard-mode)

Memory

- 512-Kbit (64-Kbyte) of EEPROM
- Page size: 128-byte
- Additional 128-byte identification page

Supply voltage

- Wide voltage range: 1.6 V to 5.5 V

Temperature

- Operating temperature range: -40 °C to +85 °C

Fast write cycle time

- Byte and page write within 4 ms (typically 3.1 ms)

Performance

- Enhanced ESD/latch-up protection
- More than 4 million write cycles
- More than 200-year data retention
- Fast wake-up time (less than 5 µs)

Ultralow power current consumption

- 330 nA (typical) in Standby mode
- 100 µA (typical) for read current
- 360 µA (typical) for write current

Advanced features

- Configurable device address register
- Device type identifier register (in read-only)
- Preprogrammed device address
- Software write protection register
- Hardware write protection of the whole memory array
- Random and sequential read modes

Package

- SO8N, TSSOP8, and UFDFPN8 (ECOPACK2)
- Unseen wafer (each die is tested)

Product label



Product status

M24512E-F

1 Description

The **M24512E-F** is a 512-Kbit I²C-compatible EEPROM (electrically erasable programmable memory) organized as 64 K x 8 bits. It can operate with a supply voltage from 1.6 V to 5.5 V with a clock frequency up to 1 MHz, over an ambient temperature range of -40 °C to +85 °C.

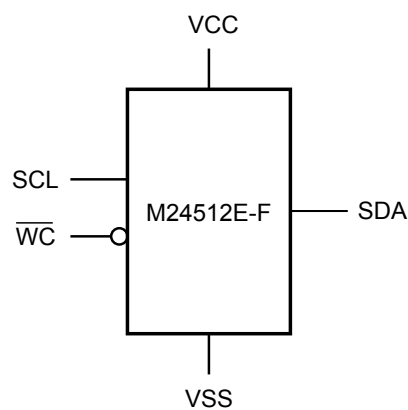
The device offers three 8-bit registers, namely: the device type identifier (DTI) register, the configurable device address (CDA) register, and the software write protection (SWP) register.

The device offers an additional 128-byte page, called the identification page. This page can be used to store sensitive application parameters, which can later be permanently locked in read-only mode.

The M24512E-F is available in the standard 8-pin TSSOP8, SO8N, UDFPN8, and unsawn wafer.

On demand, the EEPROM can be delivered with a preprogrammed and locked device address.

Figure 1. Logic diagram

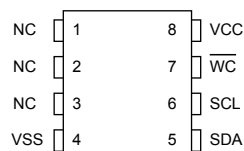


DT73083V1

Table 1. Signal names

Signal name	Description	Direction
SDA	Serial data	Input / Output
SCL	Serial clock	Input
$\overline{WC}$	Write control	Input
V _{CC}	Supply voltage	-
V _{SS}	Ground	-

Figure 2. 8-pin package connections, top view



DT54532V1

1. NC: Not connected.
2. See [Section 10: Package information](#) for package dimensions, and how to identify pin 1.

2 Signal description

2.1 Serial clock (SCL)

SCL is an input. The signal applied on it is used to strobe the data available on SDA(in) and to output the data on SDA(out).

2.2 Serial data (SDA)

SDA is an input/output used to transfer data in or out of the device. SDA(out) is an open-drain output that can be wired-AND with other open-drain or open-collector signals on the bus. A pull-up resistor must be connected from serial data (SDA) to V_{CC} (Figure 24 and Figure 25 indicate how to calculate the value of the pull-up resistor).

2.3 Write control ($\overline{WC}$)

This input signal is useful for protecting the whole content of the memory from inadvertent write operations. Write operations are:

- Disabled to the whole memory array when write control is driven high.
- Enabled when write control is either driven low or left floating.

When the write control signal is driven high, the device select and address bytes are acknowledged, but data bytes are not acknowledged.

2.4 V_{SS} (ground)

V_{SS} is the reference for the V_{CC} supply voltage.

2.5 Supply voltage (V_{CC})

2.5.1 Operating supply voltage (V_{CC})

Before selecting the memory and issuing instructions to it, a valid and stable V_{CC} voltage within the specified [$V_{CC}(\min)$, $V_{CC}(\max)$] range must be applied (see operating conditions in [Section 9: DC and AC parameters](#)).

To secure a stable DC supply voltage, it is recommended to decouple the V_{CC} line with a suitable capacitor (usually from 10 nF to 100 nF) close to the V_{CC}/V_{SS} package pins.

This voltage must remain stable and valid until the end of the transmission of the instruction and, for a write instruction, until the completion of the internal write cycle (t_W).

2.5.2 Power-up conditions

The V_{CC} voltage increases from 0 V up to the minimum V_{CC} operating voltage (see operating conditions in [Section 9](#)). Once V_{CC} is greater than or equal to the minimum V_{CC} level, the controller must wait for at least t_{WU} before sending the first command to the device. See [Table 19. AC characteristics in Fast-mode](#) and [Table 20. AC characteristics in Fast-mode Plus](#) for the value of the wake-up time parameter.

2.5.3 Device reset

To prevent inadvertent write operations during power-up, a power-on-reset (POR) circuit is included.

At power-up, the device does not respond to any instruction until V_{CC} has reached the internal reset threshold voltage. This threshold is lower than the minimum V_{CC} operating voltage (see operating conditions in [Section 9: DC and AC parameters](#)). When V_{CC} passes over the POR threshold, the device is reset and enters the standby power mode. The device must not be accessed until V_{CC} reaches a valid and stable DC voltage within the specified [$V_{CC}(\min)$, $V_{CC}(\max)$] range (see operating conditions in [Section 9: DC and AC parameters](#)).

Similarly, during power-down, when the V_{CC} decreases, the device must not be accessed once V_{CC} drops below $V_{CC}(\min)$. When V_{CC} drops below the power-on-reset threshold voltage, the device stops responding to any instruction sent to it.

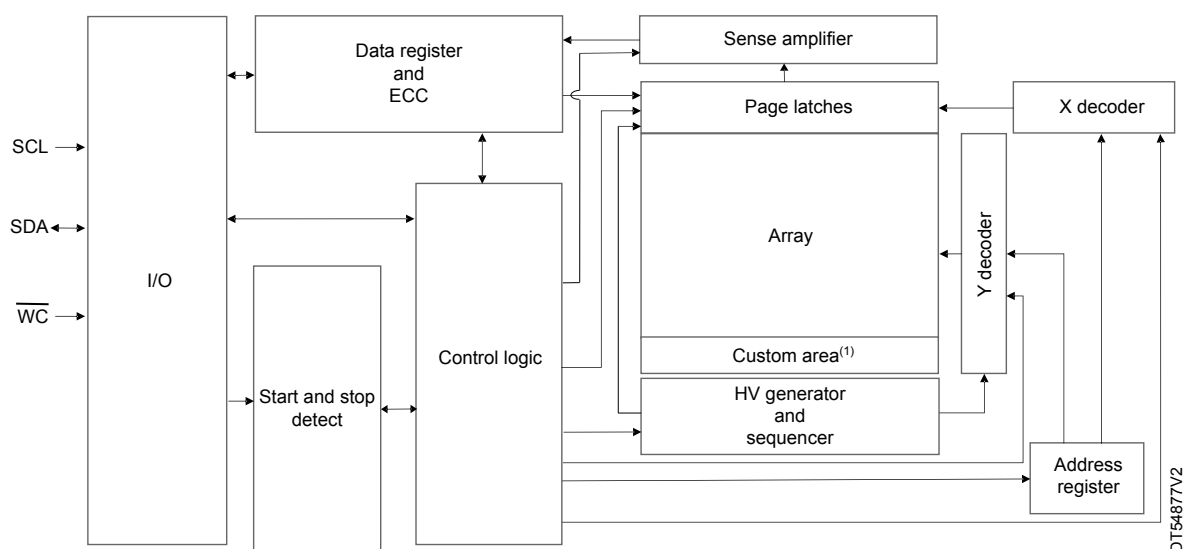
2.5.4 Power-down conditions

During power-down, when the V_{CC} decreases to 0, the device must be in the standby power mode (the mode is reached after decoding a stop condition, with no internal write cycle in progress).

3 Memory organization

The memory is organized as shown below.

Figure 3. Block diagram



1. DTI, CDA, SWP registers and identification page.

4 Device features

4.1 Device type identifier register (DTI)

The DTI is an 8-bit register permanently locked in read-only mode. This register is factory programmed with device type identifier bits (DTI3, DTI2, DTI1, and DTI0) set to 1011 and with device type identifier lock bit (DTIL) set to 1 to freeze definitively the register. DTI3, DTI2, DTI1, and DTI0 define the device type identifier address in the device select code. At power-up, the device loads the last configuration of DTI3, DTI2, DTI1, DTI0, and DTIL values.

This register is read by issuing the read device type identifier instruction. This instruction uses the same protocol and format as the random address read (from the memory array) except for the following differences (refer to Table 9, Table 10, and Table 11).

- Device type identifier = 1011
- MSB address bits A15, A14, and A13 must be equal to 111
- MSB address bits from A12 to A8 are don't care
- LSB address bits from A7 to A0 are don't care

The description of the device type identifier register is given in Table 2.

Table 2. Device type identifier register

bit 7	bit 6	bit5	bit4	bit3	bit2	bit1	bit0
DTI3 = 1	DTI2 = 0	DTI1 = 1	DTI0 = 1	X ⁽¹⁾	X	X	DTIL = 1

1. X = Don't care bit. Read as 0.

Note: The factory default value is 10110001.

Table 3. Device type identifier register description

Bit	Function
b7 to b4	DTI3, DTI2, DTI1, DTI0: Device type identifier bits. b7, b6, b5, b4 are used to configure the device type identifier of the device select code. • (b7, b6, b5, b4) = (1, 0, 1, 1) the device type identifier is 1011 (factory default value) Note: Bits b7 to b4 are frozen at factory delivery.
b3 to b1	Don't care bits. Read as 0. (b3, b2, b1) = (0, 0, 0)
b0	DTIL: Device type identifier lock bit. b0 indicates that the DTI register status is in read-only mode. • b0 = 1 the device type identifier lock bit is equal to 1 (factory default value) Note: Bit b0 is frozen at factory delivery.

4.2 Configurable device address register (CDA)

The CDA is an 8-bit register that enables the user to define a configurable device address (C2, C1, and C0), and includes a specific bit, named device address lock (DAL), to freeze definitively the configurable device address register. This register can be read and written by issuing the read or write configurable device address instruction. These instructions use the same protocol and format as the random address read or page write (from/into memory array) except for the following differences (refer to [Table 9](#), [Table 10](#), and [Table 11](#)):

- Device type identifier = 1011
- MSB address bits A15, A14, and A13 must be equal to 110
- MSB address bits from A12 to A8 are don't care
- LSB address bits from A7 to A0 are don't care

C2, C1, C0, and DAL are defining the chip enable address in the device select code and the device address lock. These bits can be written and reconfigured with a write command.

At power-up or after reprogramming, the device loads the last configuration of C2, C1, C0, and DAL values. To prevent unwanted change of configurable device address bits, the M24512E-F protects the CDA register permanently freezing it in read-only mode. The update of the CDA register is disabled (read-only) when the DAL bit is set to 1 (DAL = 1).

In the same way, the update of the CDA register is enabled when the DAL bit is set to 0 (DAL = 0). Sending more than one byte during a write configurable device address command aborts the write cycle (CDA register content does not change).

- Note:**
- *Updating the DAL bit from 0 to 1 is an irreversible action: the C2, C1, and C0, and DAL bits cannot be updated anymore.*
 - *If the write control input ($\overline{WC}$) is driven high, or if the DAL bit is set to 1 the write configurable device address command is not executed, the accompanying data byte is not acknowledged, as shown in [Figure 6](#), and the write cycle does not start.*

The description of the configurable device address register is given in [Table 4](#).

Table 4. Configurable device address register

bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
X ⁽¹⁾	X	X	X	C2	C1	C0	DAL

1. X = Don't care bits. Read as 0.

- Note:** The factory default value is 00000000, and it is 00000001 for an unsawn wafer. For further information see [Section 7: Initial delivery state](#).

For devices delivered with preprogrammed device address:

- The default value of C2, C1, and C0 are given in [Table 6](#).
- DAL is set to 1.

Table 5. Configurable device address register description

Bit	Function
b7 to b4	Don't care bits - Read as '0'. (b7, b6, b5, b4) = (0, 0, 0, 0)
b3 to b1	C2, C1, C0: Configurable device address bits. b3, b2, b1 are used to configure up to eight possibilities of chip enable address: (b3, b2, b1) = (0, 0, 0): the chip enable address is 000 (b3, b2, b1) = (0, 0, 1): the chip enable address is 001 (b3, b2, b1) = (0, 1, 0): the chip enable address is 010 (b3, b2, b1) = (0, 1, 1): the chip enable address is 011 (b3, b2, b1) = (1, 0, 0): the chip enable address is 100 (b3, b2, b1) = (1, 0, 1): the chip enable address is 101 (b3, b2, b1) = (1, 1, 0): the chip enable address is 110 (b3, b2, b1) = (1, 1, 1): the chip enable address is 111
b0	DAL: Device address lock bit. b0 locks the CDA register in read-only mode: - b0 = 0: bits b3, b2, b1, and b0 can be modified - b0 = 1: bits b3, b2, b1, and b0 cannot be modified and therefore the CDA register is frozen Note: Bits b3 to b0 can be updated (if b0 = 0) in the same write instruction. Setting b0 from 0 to 1 is an irreversible action.

If the M24512E-F is delivered with the preprogrammed device address, the configurable device address register becomes locked at factory delivery and can only be read. The C2, C1, and C0 are set as specified in Table 6. Preprogrammed device address, and the DAL bit is set to 1. The corresponding commercial product number with the preprogrammed device address are given in Table 6.

Table 6. Preprogrammed device address

Commercial product number ⁽¹⁾	Chip enable address bits			Availability ⁽²⁾
	C2	C1	C0	
M24512E-Fxx6T1	0	0	1	Yes
M24512E-Fxx6T2	0	1	0	On demand
M24512E-Fxx6T3	0	1	1	On demand
M24512E-Fxx6T4	1	0	0	Yes
M24512E-Fxx6T5	1	0	1	On demand
M24512E-Fxx6T6	1	1	0	On demand
M24512E-Fxx6T7	1	1	1	On demand

1. 'xx' in the part number varies by package.

2. The product might not be available in every package. For more information, contact the nearest ST sales office.

4.3

Software write protection register (SWP)

The software write protection (SWP) register is a nonvolatile 8-bit register that enables the user to protect a specific area of the memory against the write instruction. The SWP offers four nonvolatile bits to configure by the user:

- Two bits for setting the size of the write-protected memory and identified as block protection bits (BP0, BP1)
- One bit to enable or disable the write protection of the desired area and identified as write protect activation (WPA) bit
- One bit to definitively freeze in read-only mode the SWP register and identified as write protection lock (WPL) bit

This register can be read and written by issuing the read or write software write protection register instructions. These instructions use the same protocol and format as the random address read or page write (from/into memory array) except for the following differences (refer to Table 9, Table 10, and Table 11):

- Device type identifier = 1011
- MSB address bits A15, A14, and A13 must be equal to 101
- MSB address bits from A12 to A8 are don't care
- LSB address bits from A7 to A0 are don't care

BP1 and BP0 are the block protection bits. WPL is the write protect lock bit and WPA is the write protect activation bit. These bits can be written and reconfigured with a write command. At power-up, the device loads the last configuration of the SWP register value.

The user can update the SWP register as often as the WPL bit stays at 0. Writing more than one byte discard the write cycle (software write protection register content is not changed).

To prevent unwanted change of software write-protection register bits, the M24512E-F protects the SWP register, freezing it permanently in read-only mode. The update of the SWP register is disabled (read only) when the WPL bit is set to 1 (WPL = 1). In the same way, the update of the SWP register is enabled when the WPL bit is set to 0 (WPL = 0).

When WPL is set to 1 and writing software write protection register, the device select and address bytes are acknowledged, data byte is not acknowledged and the write cycle does not start.

- Note:*
- *Updating the WPL bit from 0 to 1 is an irreversible action: the WPA, BP1, BP0, and WPL bits cannot be updated any more*
 - *If the write control input ($\overline{WC}$) is driven high or if the WPL bit is set to 1, the write command on the software write protection register is not executed and the accompanying data byte is not acknowledged, as shown in Figure 10.*

The description of the software write protection register is given in the below table:

Table 7. Software write protection register values

bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
X ⁽¹⁾	X	X	X	WPA	BP1	BP0	WPL

1. X = Don't care bits. Read as 0.

- Note:* The factory default value is 00000000.

Table 8. Software write protection register description

Bit	Function
b7 to b4	• Don't care bits. Read as 0. (b7, b6, b5, b4) = (0, 0, 0, 0)
b3	WPA: Write protect activation bit. b3 enables or disables the write protection: • b3 = 0: no write protection. The whole memory can be written. • b3 = 1: write protection active. The memory block is protected according to BP bits setting.
b2 to b1	BP1, BP0: Block protection bits b2 and b1 define the size of the memory block to be protected against write instruction: • (b2, b1) = (0, 0): the upper quarter of the memory is write-protected • (b2, b1) = (0, 1): the upper half of the memory is write protected • (b2, b1) = (1, 0): the upper ¾ of the memory is write protected • (b2, b1) = (1, 1): the whole memory is write protected
b0	WPL: Write protection lock bit b0 locks the write protection register value. • b0 = 0: bits [b3: b0] can be modified • b0 = 1: bits [b3: b0] cannot be modified and therefore the write protection register is frozen. Note: bits b3 to b0 can be updated (if b0 = 0) in the same write instruction. Setting b0 from 0 to 1 is an irreversible action.

4.4 Identification page

The identification page (128 bytes) is an additional page that can be read or written and can later be permanently locked in read-only mode. It is read or written by issuing the read or write identification page instruction. These instructions use the same protocol and format as the random address read or page write (from memory array) except for the following differences (refer to [Table 9](#), [Table 10](#), and [Table 11](#)):

- Device type identifier = 1011
- MSB address bits A15, A14, and A13 must be equal to 000
- MSB address bits from A12 to A8 are don't care
- LSB address bit A7 is don't care, bits A6 to A0 define the byte address inside the identification page

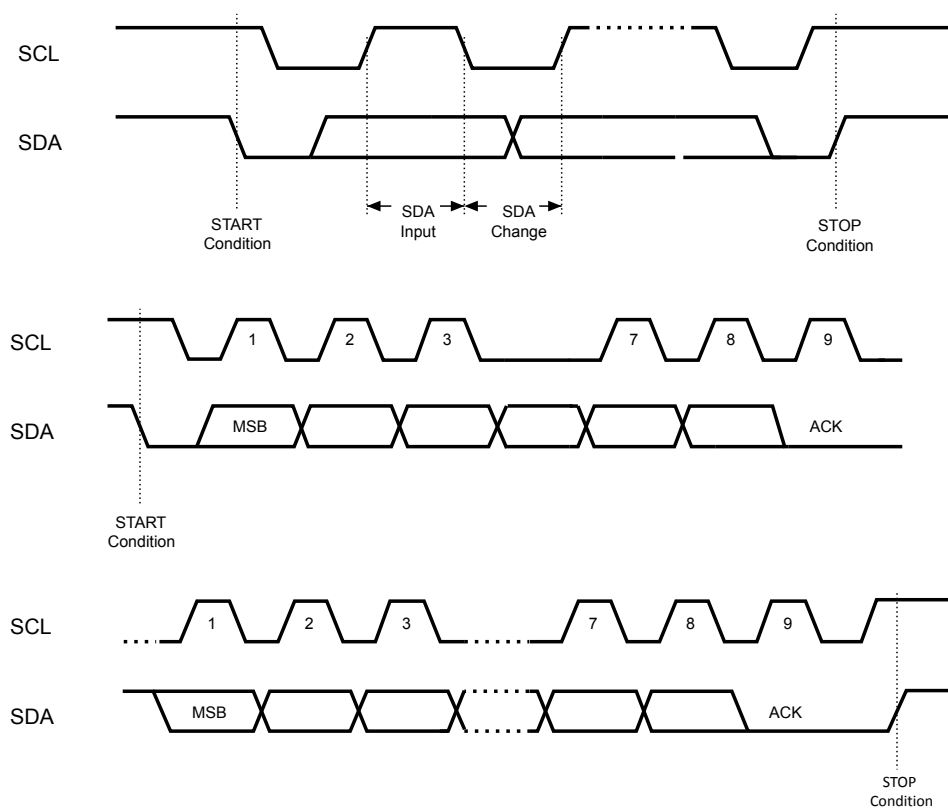
If the identification page is locked, the data bytes transferred during the write identification page instruction are not acknowledged (NO ACK).

The identification page is filled with all bytes written to FFh.

5 Device operation

The device supports the I²C protocol summarized in Figure 4. Any device that sends data onto the bus is defined as a transmitter, and any device that reads the data is defined as a receiver. The device that controls the data transfer is known as the bus controller, and the other as the target. A data transfer can only be initiated by the bus controller, which also provides the serial clock for synchronization. The device is always a target in all communications.

Figure 4. I²C bus protocol



DT50012V1

5.1 Start condition

The start condition is identified by a falling edge of serial data (SDA) while the serial clock (SCL) is stable in the high state. This condition must precede any data transfer instruction. The device continuously monitors the SDA and SCL for a start signal, except during a write cycle.

5.2 Stop condition

The stop condition is identified by a rising edge of serial data (SDA) while the serial clock (SCL) is stable in the high state. This condition terminates the communication between the device and the bus controller. A read instruction followed by NO ACK can be followed by a stop condition to force the device into the standby mode. A stop condition at the end of a write instruction triggers the internal write cycle.

5.3 Data input

During data input, the device samples the serial data (SDA) on the rising edge of the serial clock (SCL). For proper device operation, the SDA must be stable during the rising edge of the SCL, and the SDA signal must change only when the SCL is driven low.

5.4 Acknowledge bit (ACK)

The acknowledge bit is used to indicate a successful byte transfer. The bus transmitter, whether a bus controller or target device, releases serial data (SDA) after sending eight bits of data. During the ninth clock pulse period, the receiver pulls SDA low to acknowledge the receipt of the eight data bits.

5.5 Device addressing

To start communication between the bus controller and the target device, the bus controller must initiate a start condition. Following this, and unless the product is delivered with the preprogrammed device address, the bus controller sends the device select code and byte address as specified in Table 9, Table 10, and Table 11. When the device select code is received, the device responds only if the bits b3, b2 and b1 values match the values of the C2, C1 and C0 bits programmed in the configurable device address register.

If a match occurs, the corresponding device gives an acknowledgment on serial data (SDA) during the ninth bit time. If the device does not acknowledge the device select code, it deselects itself from the bus and goes into standby mode after a stop condition.

The eighth bit is the read/write bit ($\overline{RW}$). This bit is set to 1 for read operations and 0 for write operations.

Table 9. Device select code

Features	Device type identifier bits				Chip enable address bits ⁽¹⁾			$\overline{RW}$
	Bit 7 (MSB) ⁽²⁾	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSB)
Memory	1	0	1	0	C2	C1	C0	$\overline{RW}$
Device type identifier	1	0	1	1	C2	C1	C0	$\overline{RW}$
Configurable device address	1	0	1	1	C2	C1	C0	$\overline{RW}$
Software write protection	1	0	1	1	C2	C1	C0	$\overline{RW}$
Identification page	1	0	1	1	C2	C1	C0	$\overline{RW}$
Identification page lock	1	0	1	1	C2	C1	C0	$\overline{RW}$

1. C0, C1 and C2 are compared with the value read on bits b1, b2, and b3 of the CDA register.

2. The most significant bit, b7, is sent first.

Table 10. First byte address

Features	Bit 7 (MSB) ⁽¹⁾	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSB)
Memory	A15	A14	A13	A12	A11	A10	A9	A8
Device type identifier	1	1	1	X ⁽²⁾	X	X	X	X
Configurable device address	1	1	0	X	X	X	X	X
Software write protection	1	0	1	X	X	X	X	X
Identification page	0	0	0	X	X	X	X	X
Identification page lock	0	1	1	X	X	X	X	X

1. The most significant bit, b7, is sent first.

2. X = Don't care bit.

Table 11. Second byte address

Features	Bit 7 (MSB) ⁽¹⁾	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSB)
Memory	A7	A6	A5	A4	A3	A2	A1	A0
Device type identifier	X ⁽²⁾	X	X	X	X	X	X	X
Configurable device address	X	X	X	X	X	X	X	X
Software write protection	X	X	X	X	X	X	X	X
Identification page	X	A6	A5	A4	A3	A2	A1	A0
Identification page lock	X	X	X	X	X	X	X	X

1. The most significant bit, b7, is sent first.

2. X = Don't care bit

6 Instructions

6.1 Write operations on memory array

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) reset to 0. The device acknowledges this, as shown in [Figure 5](#), and waits for two address bytes. The device responds to each address byte with an acknowledge bit, and then waits for the data byte. See in [Section 5.5: Device addressing](#) ([Table 9](#), [Table 10](#), and [Table 11](#)) how to address the memory array.

When the bus controller generates a stop condition immediately after a data byte ACK bit (in the tenth bit time slot), either at the end of a byte write or a page write, the internal write cycle t_W is triggered. A stop condition at any other time slot does not trigger the internal write cycle.

After the stop condition and the successful completion of an internal write cycle (t_W), the device internal address counter is automatically incremented to point to the next byte after the last modified byte.

During the internal write cycle, serial data (SDA) is disabled internally, and the device does not respond to any requests.

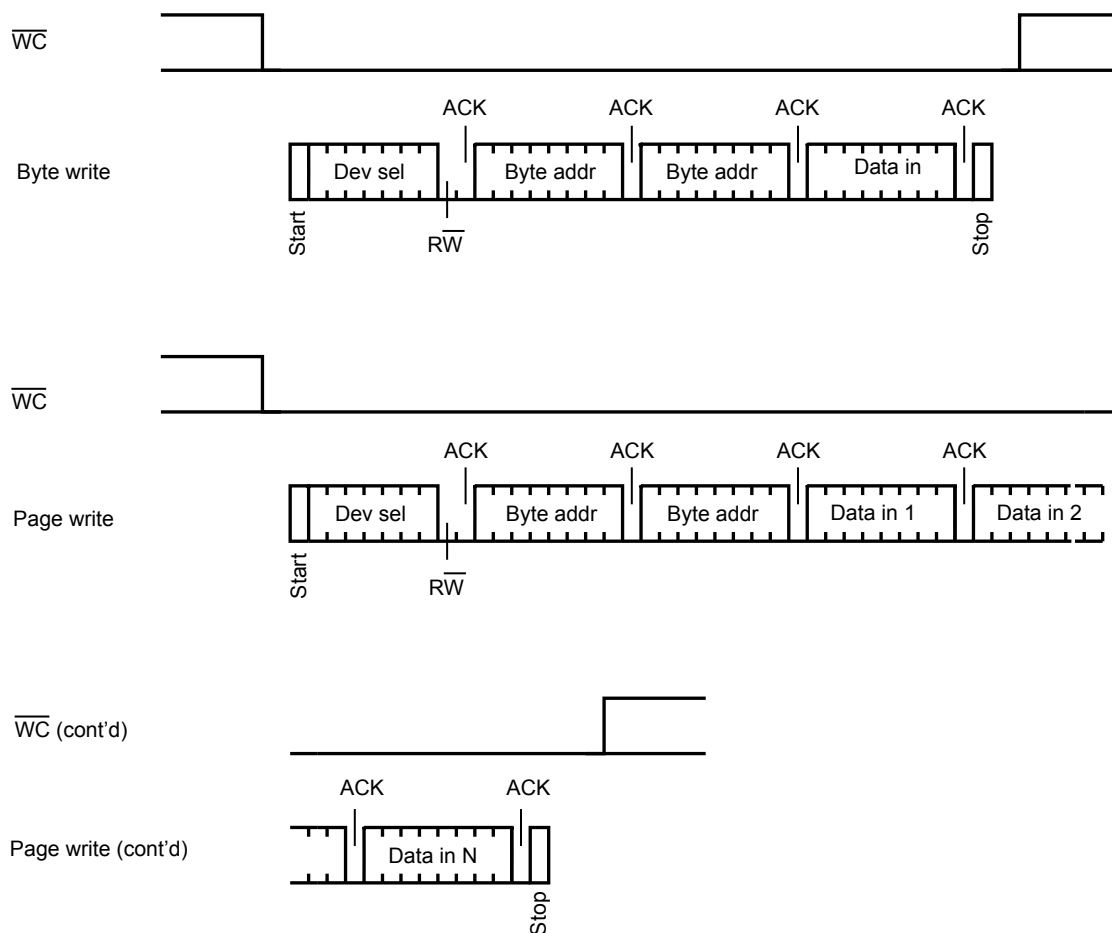
If the addressed area is write protected by software through the SWP setting or hard protected through $\overline{WC}$ pin driven high, the write instruction is not executed and the accompanying data bytes are not acknowledged (as shown in [Figure 6](#)).

6.1.1

Byte write

After the device select code and the address bytes, the bus controller sends one data byte. If the addressed location is write-protected, through the SWP setting or through the $\overline{WC}$ pin being driven high, the device replies with NO ACK, and the location is not modified. If the addressed location is not write-protected, the device replies with ACK. The bus controller terminates the transfer by generating a stop condition, as shown in Figure 5.

Figure 5. Write mode sequences without write protection (data write enabled)



DT01106dV2

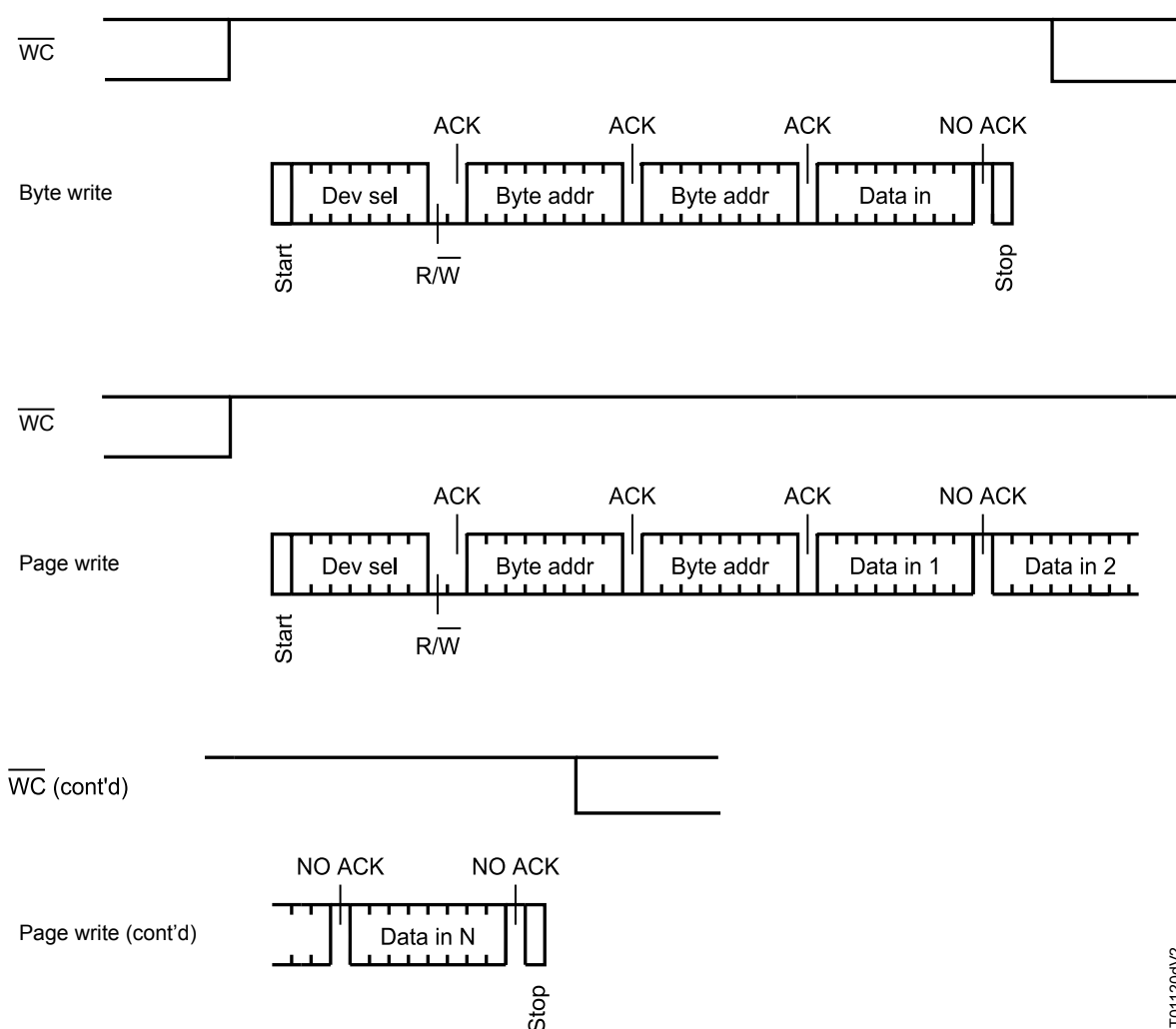
6.1.2 Page write

Using this mode, it is possible to write up to 128 bytes in a single write cycle, provided they are all located on the same page. This means that the most significant memory address bits, from A15 to A7, are the same. If more bytes are sent than fit up to the end of the page, a rollover occurs: the bytes exceeding the page end are written on the same page, starting from location 0.

The bus controller sends from 1 to 128 bytes of data, each of which is acknowledged by the device if the addressed bytes are not write-protected through the SWP setting or through $\overline{WC}$ pin (driven low). In the opposite case, when the addressed bytes are write-protected by the SWP setting or $\overline{WC}$ pin (driven high), the contents of the addressed memory location are not modified, and each data byte is followed by a NO ACK, as shown in Figure 6. After each transferred byte, the internal page address counter is incremented.

The transfer is terminated when the bus controller generates a stop condition.

Figure 6. Write mode sequences with write protection (data write inhibited)



DT01120dV2

6.2 Write operations on features

6.2.1 Write operation on DTI register

Write operations on the device type identifier (DTI) register are not allowed.

The register is delivered locked in read-only.

6.2.2 Write operation on CDA register

Write operations on the configurable device address register are performed according to the state of the device address lock bit (DAL) or the status of the $\overline{WC}$ line.

If the configurable device address register is write protected by software with $DAL = 1$ or hard protected with $\overline{WC}$ line driven high, the write operation on this register is not executed and the accompanying data byte is not acknowledged as shown in Figure 8.

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this, as shown in Figure 7, and waits for the address bytes where the register is located. The device responds to each address byte with an acknowledge bit, and then waits for the data byte. See in Table 9, Table 10, and Table 11 how to address the configurable device address register.

When the bus controller generates a stop condition immediately after the data byte ACK bit (in the tenth bit time slot), the internal write cycle t_W is triggered. A stop condition at any other time slot does not trigger the internal write cycle.

During the internal write cycle, serial data (SDA) is disabled internally, and the device does not respond to any requests (NO ACK).

If the three bits C2, C1, and C0 have been reconfigured with a correct write command, the device acknowledges if the chip enable address of the device select code is equal to the new values of C2, C1, and C0, otherwise NO ACK. Sending more than one byte aborts the write cycle (configurable device address content does not change). Bits (C2, C1, C0, and DAL) can be updated ($DAL = 0$ to 1) in the same program instruction.

Figure 7. Write CDA register (data write enabled)

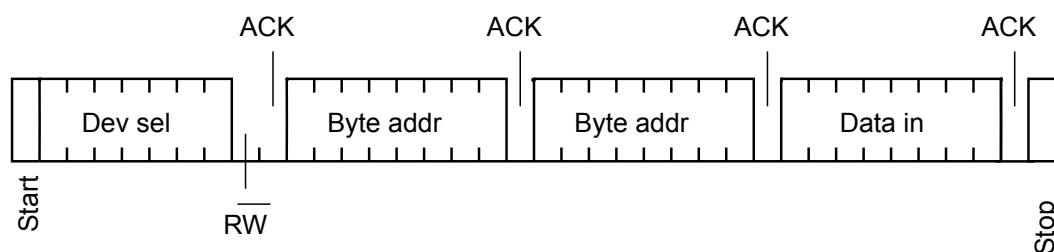
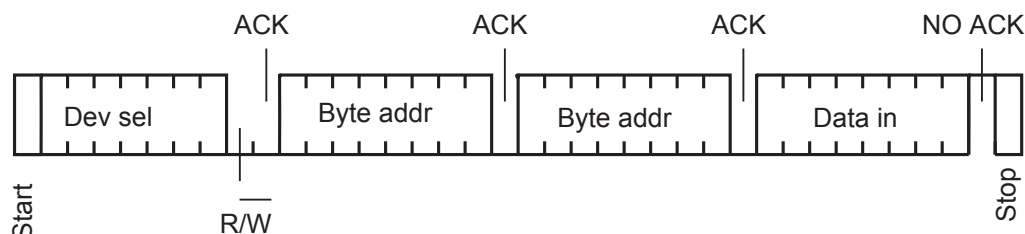


Figure 8. Write CDA register (data write inhibited by software or hardware)



6.2.3 Write operation on SWP register

Write operations on SWP register are performed according to the state of the write protect lock bit (WPL) or to the status of the $\overline{WC}$ line.

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this, as shown in Figure 9, and waits for the address bytes where the SWP register is located. The device responds to each address byte with an acknowledge bit, and then waits for the data byte. See in Table 9, Table 10, and Table 11 how to address the software write protection register.

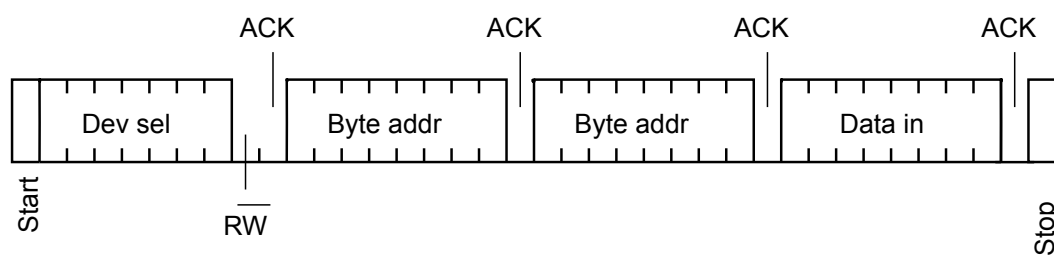
When the bus controller generates a stop condition immediately after the data byte ACK bit (in the tenth bit time slot), the internal write cycle t_W is triggered. A stop condition at any other time slot does not trigger the internal write cycle.

During the internal write cycle, serial data (SDA) is disabled internally, and the device does not respond to any requests (NO ACK).

Sending more than one byte aborts the write cycle. The software write protection register content does not change.

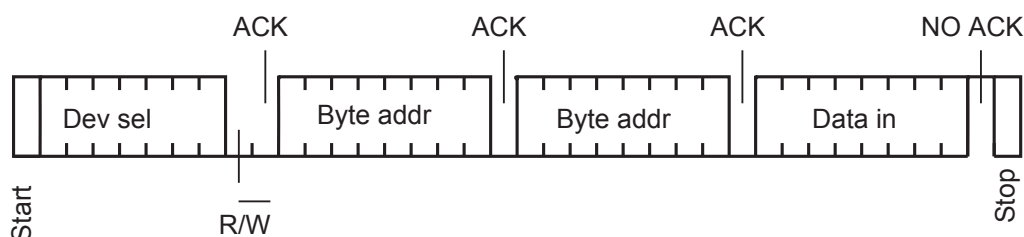
If the SWP register is already locked or hard write protected with $\overline{WC}$ line driven high, the write operation is not executed and the accompanying data byte is not acknowledged as shown in Figure 10.

Figure 9. Write SWP register (data write enabled)



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Figure 10. Write SWP register (data write inhibited by software or hardware)



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6.2.4 Write operation on identification page

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this, as shown in Figure 11, and waits for the address bytes where the identification page is located. The device responds to each address byte with an acknowledge bit, and then waits for the data byte. See in Section 5.5: Device addressing (Table 9, Table 10 and Table 11) how to address the identification page.

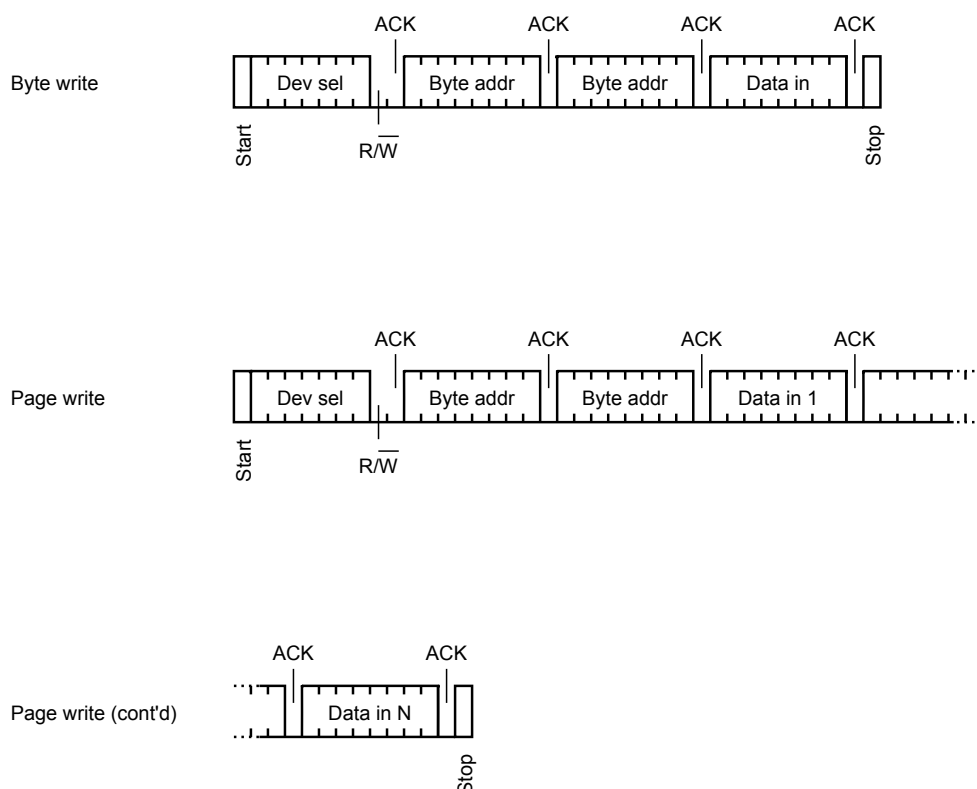
When the bus controller generates a stop condition immediately after the data byte ACK bit (in the tenth bit time slot), the internal write cycle t_W is triggered. The device internal address counter is automatically incremented to point to the next byte after the last modified byte.

A stop condition at any other time slot does not trigger the internal write cycle.

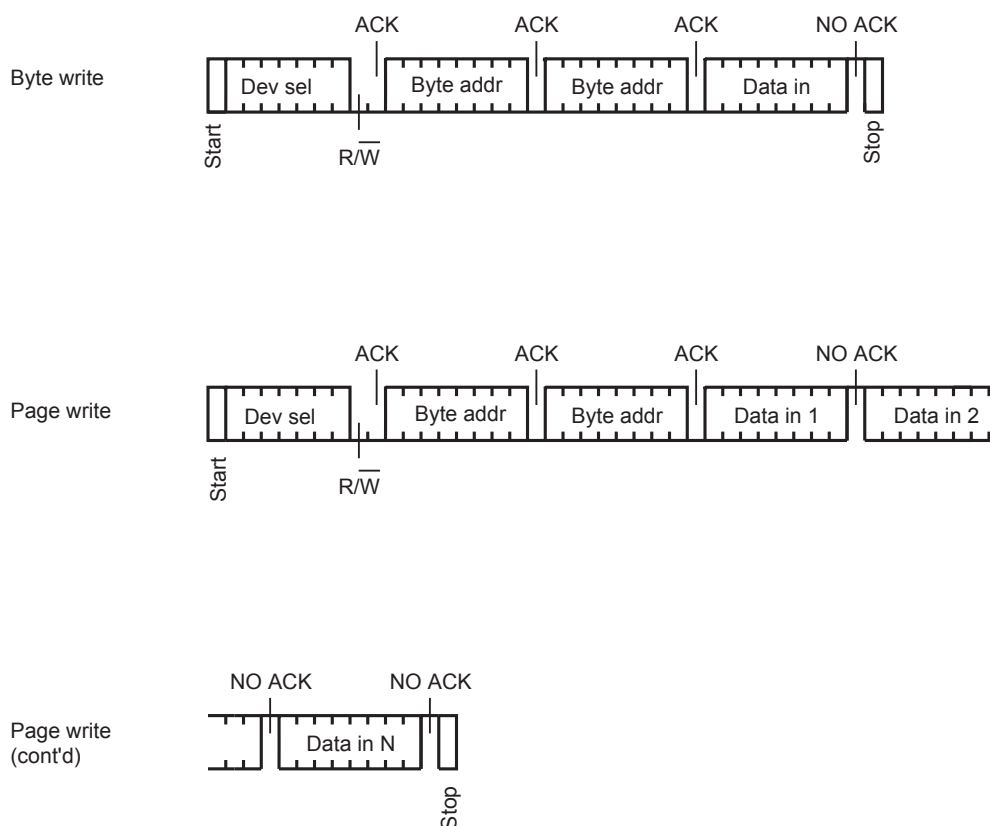
During the internal write cycle, serial data (SDA) is disabled internally, and the device does not respond to any requests (NO ACK).

If the identification page is already locked or hard write protected with $\overline{WC}$ line driven high, the write operation is not executed and the accompanying data bytes are not acknowledged as shown in Figure 12.

Figure 11. Write identification page (page unlocked)



DT01106v1

Figure 12. Write identification page (page locked or hard protected)


DT01120eV1

6.2.5 Lock operation on identification page

The lock identification page instruction (lock ID) permanently locks the identification page in read-only mode.

After a start condition, the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this, as shown in Figure 13, and waits for the address bytes where the identification page is located. The device responds to each address byte with an acknowledge bit, and then waits for a specific data byte value. See Table 9, Table 10, and Table 11 for details on addressing the identification page.

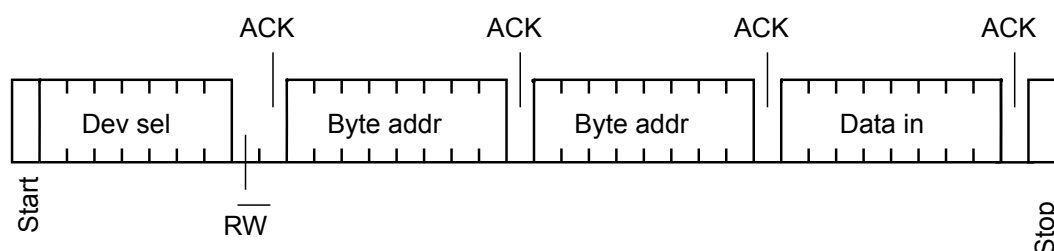
The data byte must be equal to the binary value $xxxx\ xx1x$, where x is a "don't care" bit.

When the bus controller generates a stop condition immediately after the data byte ACK bit (in the tenth bit time slot), the internal write cycle t_W is triggered. A stop condition at any other time slot does not trigger the internal write cycle.

During the internal write cycle, serial data (SDA) is disabled internally, and the device does not respond to any requests (NO ACK).

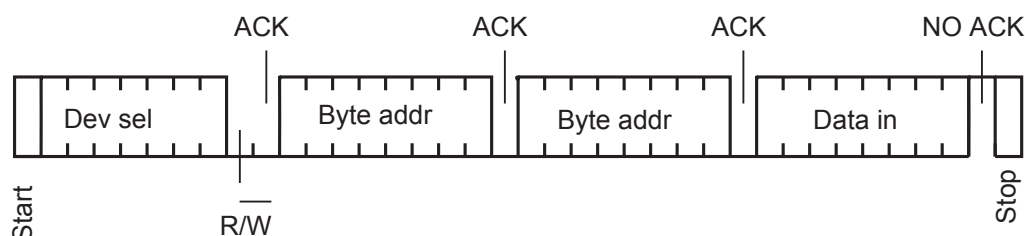
If the identification page is already locked or hard write protected with $\overline{WC}$ line driven high, the write operation is not executed and the accompanying data bytes are not acknowledged as shown in Figure 14.

Figure 13. Lock operation on identification page (unlocked or data write enabled)



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Figure 14. Lock operation on identification page (already locked or data write inhibited by hardware)



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6.3 Minimizing write delays by polling on ACK

During the internal write cycle, the device disconnects itself from the bus, and writes a copy of the data from its internal latches to the memory cells. The maximum write time (t_w) is shown in [Table 19. AC characteristics in Fast-mode](#) and [Table 20. AC characteristics in Fast-mode Plus](#), but the typical time is shorter. The bus controller can implement a polling sequence to utilize this feature.

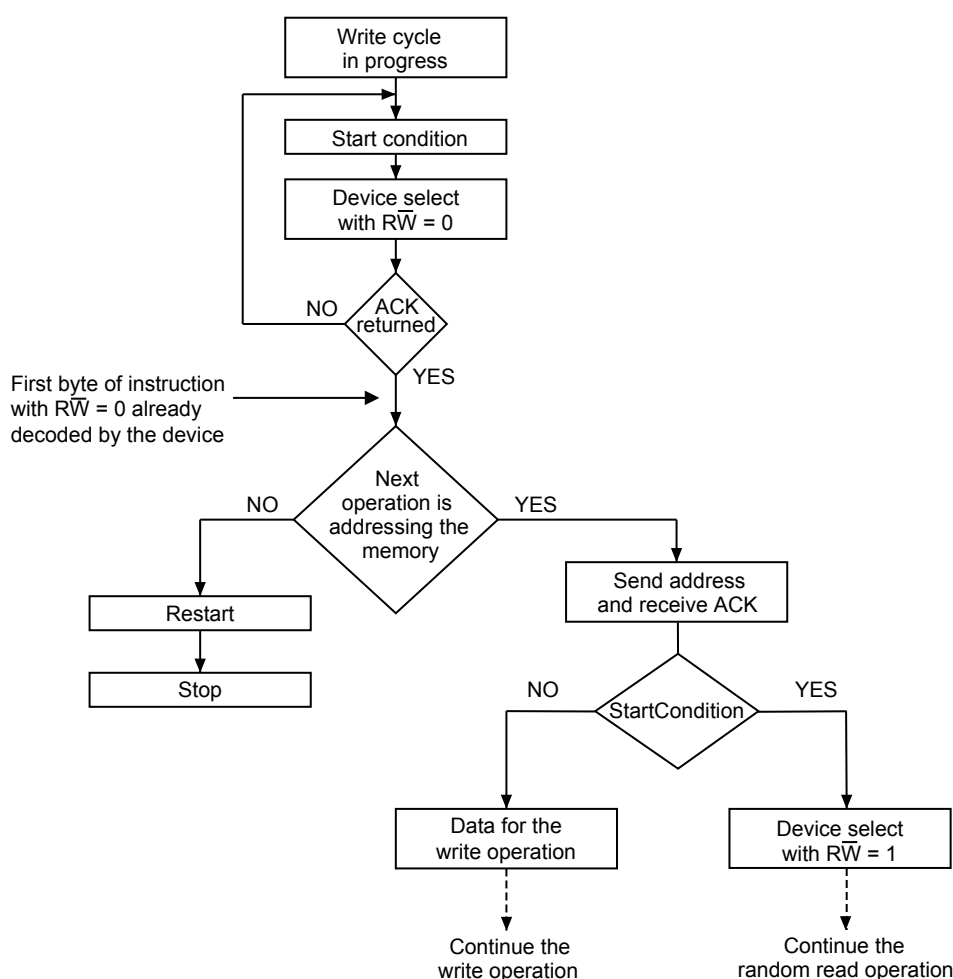
The sequence, as shown in [Figure 15](#), is:

- Initial condition: A write cycle is in progress.
- Step 1: The bus controller issues a Start condition followed by a device select code (the first byte of the new instruction).
- Step 2: If the device is busy with the internal write cycle, NO ACK is returned and the bus controller goes back to step 1. If the device has terminated the internal write cycle, it responds with an ACK, indicating that the device is ready to receive the second part of the instruction (the first byte of this instruction having been sent during step 1).

Note: When updating the configurable device address register with C2, C1, and C0 reconfigured, the device returns ACK only if:

- The chip enable address of the device select code is equal to the new C2, C1, and C0 values.
- An internal write cycle is completed (a new C2, C1, and C0 values have been programmed in the chip enable register).

Figure 15. Write cycle polling flowchart using ACK



Note: The seven most significant bits of the device select code in a random read (bottom right box in the figure above) must match those of the device select code in the write operation (polling instruction).

6.4 ECC (error correction code) and write cycling

ECC is an internal logic function transparent for the I²C communication protocol.

The ECC logic is implemented on each group of four bytes (located at addresses $[4*N, 4*N+1, 4*N+2, 4*N+3]$, where N is an integer). Within a group, if a single bit happens to be erroneous during a read operation, the ECC detects and replaces it with the correct value. The read reliability is therefore much improved.

Even if the ECC function is performed on groups of four bytes, a single byte can be written/cycled independently. In this case, the ECC function also writes/cycles the three other bytes located in the same group. As a consequence, the maximum cycling budget is defined at group level and the cycling can be distributed over the four bytes of the group: the sum of the cycles seen by byte0, byte1, byte2, and byte3 of the same group must remain below the maximum value defined in [Table 16](#).

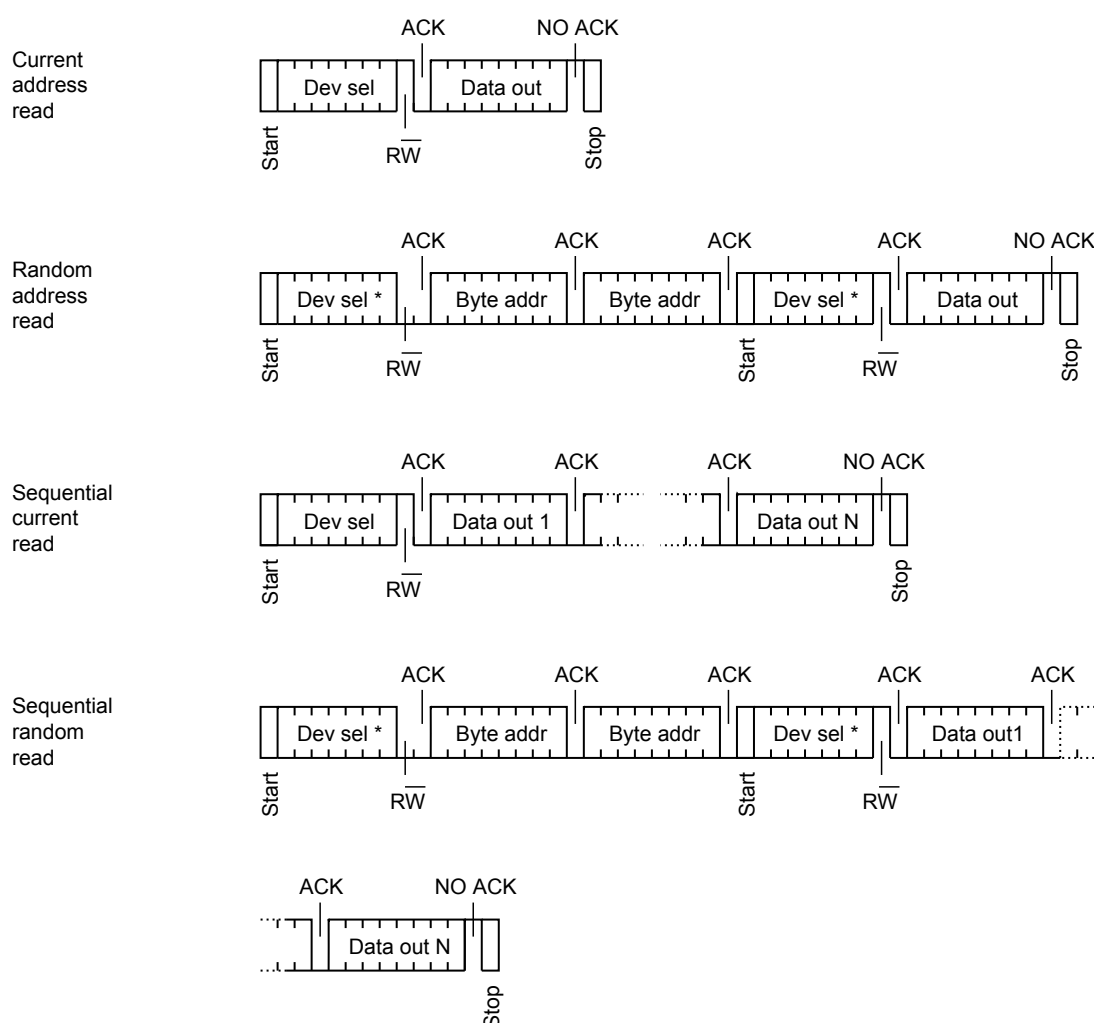
6.5 Read operations on the memory array

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the two-byte address. The device responds to each address byte with an acknowledge bit. Then, the bus controller sends another start condition, and repeats the device select code, with the $\overline{RW}$ bit set to 1. The device acknowledges this, and outputs the contents of the data. See [Table 9](#), [Table 10](#), and [Table 11](#) for details on addressing the memory array.

After each byte read (data out), the device waits for an acknowledgment (data in) during the ninth bit time. If the bus controller does not acknowledge during this interval, the device terminates the data transfer and switches to its standby mode after a stop condition.

After the successful completion of a read operation, the internal address counter is incremented by one, to point to the next byte address.

Figure 16. Read mode sequences



Note: The seven most significant bits of the first device select code in a random read must match those of the device select code in the write operation.

6.5.1 Random address read

A dummy write is first performed to load the address into this address counter (as shown in [Figure 16](#)) but without sending a stop condition. Then, the bus controller sends another start condition, and repeats the device select code, with the RW bit set to 1. The device acknowledges this, and outputs the contents of the addressed byte. The bus controller must not acknowledge the byte, and terminates the transfer with a stop condition.

6.5.2 Current address read

For the current address read operation, following a start condition, the bus controller sends only a device select code with the RW bit set to 1. The device acknowledges this and outputs the byte addressed by the internal address counter. The counter is then incremented. The bus controller terminates the transfer with a stop condition, as shown in [Figure 16](#), without acknowledging the byte.

Note: The address counter value is defined by instructions accessing either the memory, the registers, or the identification page. When accessing the registers or the identification page, the address counter value is loaded with the byte location, therefore the next current address read in the memory uses this new address counter value. When accessing the memory, it is safer to always use the random address read instruction (this instruction loads the address counter with the byte location to read in the memory, see [Section 6.5.1: Random address read](#)) instead of the current address read instruction.

6.5.3 Sequential read

This operation can be used after a current address read or a random address read. The bus controller does acknowledge the data byte output and sends additional clock pulses so that the device continues to output the next byte in sequence. To terminate the stream of bytes, the bus controller must not acknowledge the last byte, and must generate a stop condition, as shown in [Figure 16](#).

The output data comes from consecutive addresses, with the internal address counter automatically incremented after each byte is output. After reaching the last memory address, the address counter rolls over, and the device continues to output data from the memory address 00h.

6.6 Read operations on features

Only the random address read or sequential random read commands are authorized to access the DTI, SWP, and CDA registers. The address counter contains a meaningful address value only after these authorized commands have been performed.

6.6.1 Read operation on DTI register

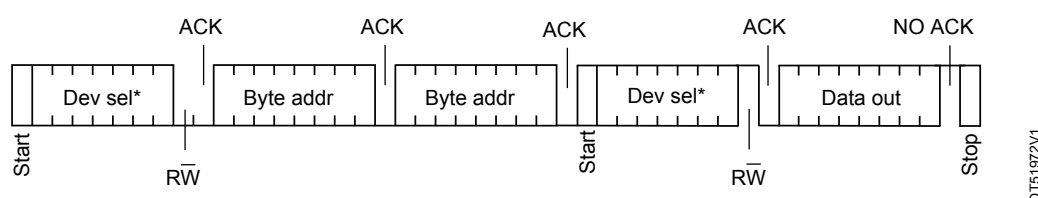
Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the address bytes where the DTI register is located. The device responds to each address byte with an acknowledge bit. Then, the bus controller sends another start condition, and repeats the device select code, with the $\overline{RW}$ bit set to 1. The device acknowledges this, and outputs the contents of the DTI register. See Table 9, Table 10, and Table 11 for details on addressing the device type identifier register.

After the successful completion of a read device type identifier, the device internal address counter is not incremented by one, to point to the next byte address. Reading more than one byte with the sequential random read command loops on reading the device type identifier register value.

To terminate the stream of data byte, the bus controller must not acknowledge the byte, and must generate a stop condition, as shown in Figure 17.

The device type identifier register cannot be read while a write cycle (t_W) is ongoing.

Figure 17. Random read DTI register



Note: The seven most significant bits of the first device select code in a random read must match those of the device select code in the write operation.

6.6.2 Read operation on CDA register

Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the address bytes where the CDA register is located. The device responds to each address byte with an acknowledge bit. Then, the bus controller sends another start condition, and repeats the device select code, with the $\overline{RW}$ bit set to 1. The device acknowledges this, and outputs the contents of the CDA register. See Table 9, Table 10, and Table 11 how to address the configurable device address register.

After the successful completion of a read configurable device address, the device internal address counter is not incremented by one, to point to the next byte address. Reading more than one byte loops on reading the configurable device address register value.

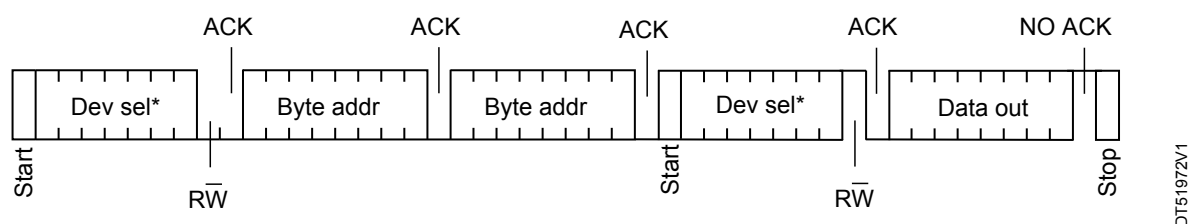
To terminate the stream of data byte, the bus controller must not acknowledge the byte, and must generate a stop condition, as shown in Figure 18.

The configurable device address register cannot be read while a write cycle (t_W) is ongoing.

The configurable device address bit (C2, C1, and C0) value can be checked by sending the device select code.

- If the chip enable address b3, b2, and b1 sent in the device select code matches the C2, C1, and C0 value, the device sends an ACK.
- Otherwise, the device answers NO ACK.

Figure 18. Random read CDA register



Note: The seven most significant bits of the first device select code in a random read must match those of the device select code in the write operation.

6.6.3 Read operation on SWP register

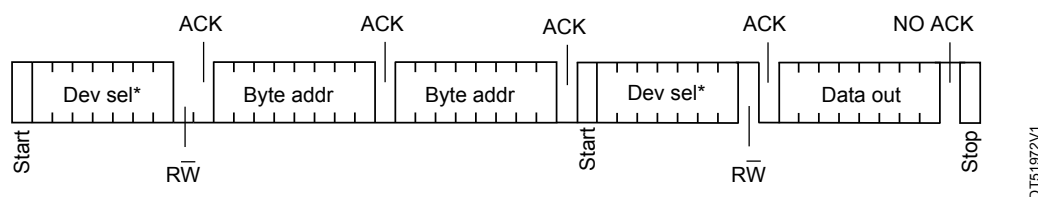
Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the address bytes where the SWP register is located. The device responds to each address byte with an acknowledge bit. Then, the bus controller sends another start condition, and repeats the device select code, with the $\overline{RW}$ bit set to 1. The device acknowledges this, and outputs the contents of the SWP register. See Table 9, Table 10, and Table 11 for details on addressing the software write protection register.

After the successful completion of a read operation on SWP, the device internal address counter is not incremented by one, to point to the next byte address. Reading more than one-byte loops on reading the SWP register value.

To terminate the stream of data byte, the bus controller must not acknowledge the byte, and must generate a stop condition, as shown in Figure 19.

The SWP register cannot be read while a write cycle (t_W) is ongoing.

Figure 19. Random read SWP register



Note: The seven most significant bits of the first device select code in a random read must match those of the device select code in the write operation.

6.6.4 Read operation on identification page

After a start condition, the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the address bytes where the identification page is located. The device responds to each address byte with an acknowledge bit. Then, the bus controller sends another start condition, and repeats the device select code, with the $\overline{RW}$ bit set to 1. The device acknowledges this, and outputs the contents of the identification page. See Table 9, Table 10, and Table 11 for details on addressing the identification page.

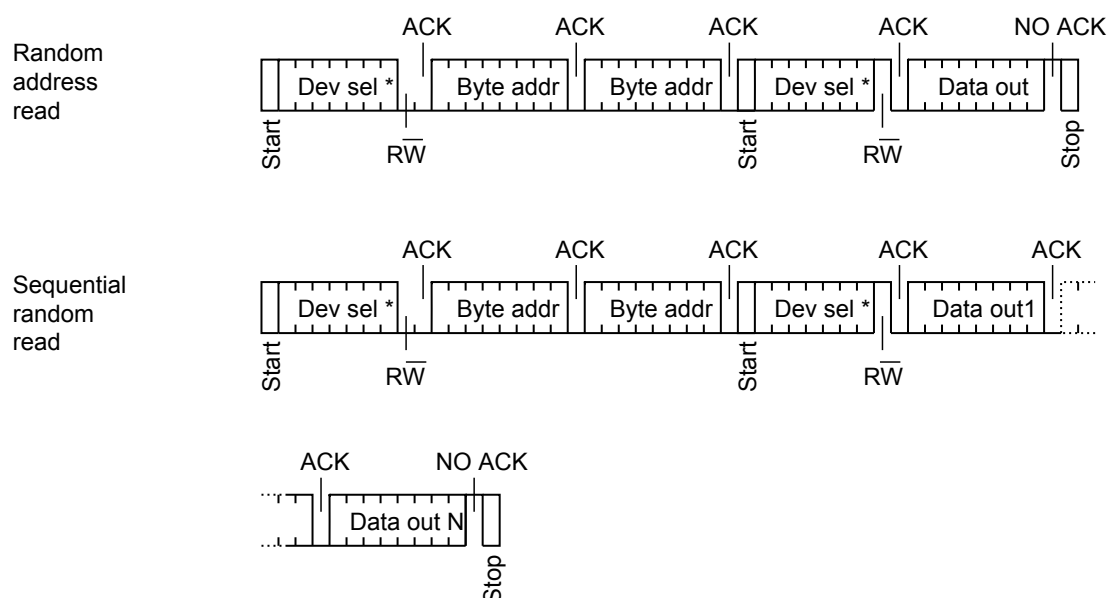
After each byte read (data out), the device waits for an acknowledgment (data in) during the ninth bit time.

The output data of the identification page comes from consecutive addresses, with the internal address counter automatically incremented after each byte output. After the last identification page address (7Fh), the address counter rolls-over, and the device continues to output data from identification page address 00h.

To terminate the stream of data byte, the bus controller must not acknowledge the byte, and must generate a stop condition, as shown in Figure 20.

If the bus controller does not acknowledge during this ninth time, the device terminates the data transfer as shown in Figure 20 and switches to its standby mode.

Figure 20. Random read identification page



Note: The seven most significant bits of the first device select code in a random read must match those of the device select code in the write operation.

6.6.5 Read lock status on identification page

The lock or unlock status of the identification page can be checked by transmitting a specific truncated command. Following a start condition the bus controller sends a device select code with the R/W bit ($\overline{RW}$) set to 0. The device acknowledges this and waits for the address bytes where the identification page is located. The device responds to each address byte with an acknowledge bit, and then waits for the data byte. See in [Table 9](#), [Table 10](#), and [Table 11](#) how to address the identification page.

The device returns an acknowledge bit after the data byte if the identification page is unlocked (unlock status) as shown in [Figure 21](#), otherwise a NO ACK bit as shown in [Figure 22](#), if the identification page is locked (lock status).

Right after this, it is recommended to transmit to the device a start condition followed by a stop condition, so that:

- Start: the truncated command is not executed because the start condition resets the device internal logic
- Stop: the device is then set back into standby mode by the stop condition

Figure 21. Read lock status (identification page unlocked)

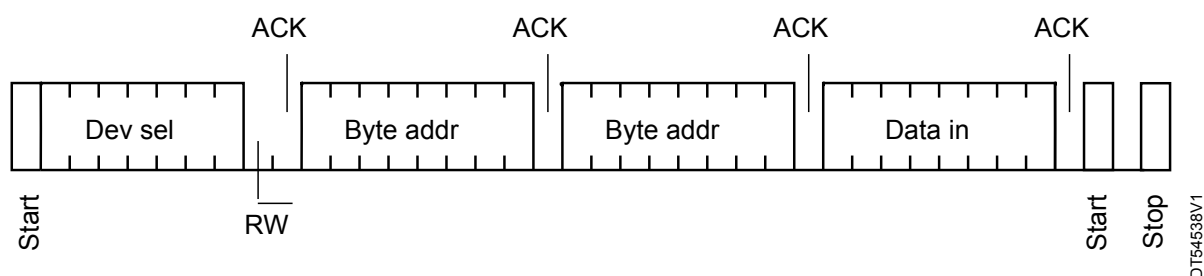
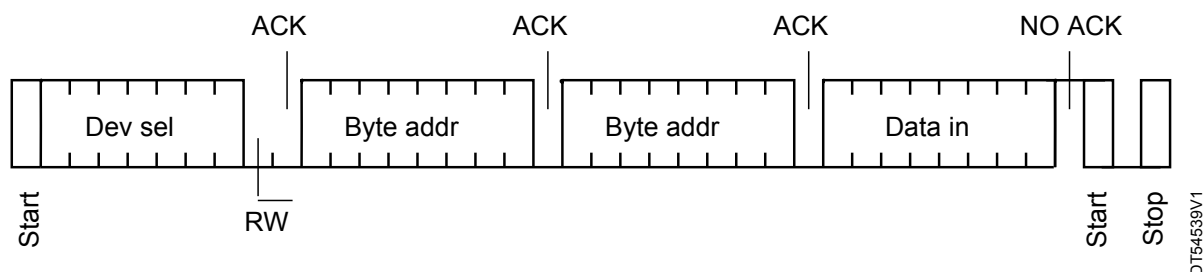


Figure 22. Read lock status (identification page locked)



7 Initial delivery state

At factory delivery, unless a device-specific address has been preprogrammed, the device is delivered with:

- All the memory array bits set to 1 (each byte contains FFh)
- The DTI register locked and set to 10110001 (B1h)
- The CDA register set to 00000000 (00h) except for unsawn wafer set to 00000001 (01h)
- The SWP register set to 00000000 (00h)
- All the identification page bits set to 1 (each byte contains FFh)

8 Maximum ratings

Stressing the device outside the ratings listed in Table 12 may permanently damage it. These are stress ratings only, and operation of the device at these, or any other conditions outside those indicated in the operating sections of this specification, is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Table 12. Absolute maximum ratings

Symbol	Parameter	Min.	Max.	Unit
-	Ambient operating temperature	-40	130	°C
T _{STG}	Storage temperature	-65	150	°C
T _{LEAD}	Lead temperature during soldering	see note ⁽¹⁾		°C
I _{OL}	DC output current (SDA = 0)	-	5	mA
V _{IO}	Input or output range	-0.50	6.5	V
V _{CC}	Supply voltage	-0.50	6.5	V
V _{ESD}	Electrostatic pulse (human body model) ⁽²⁾	-	4000	V

1. Compliant with JEDEC standard J-STD-020 (for small body, Sn-Pb or Pb-free assembly), the ST ECOPACK 7191395 specification, and the European directive on restrictions of hazardous substances (RoHS directive 2011/65/EU of July 2011).
2. Positive and negative pulses applied on different combinations of pin connections, according to ANSI/ESDA/JEDEC JS-001 (C1 = 100 pF, R1 = 1500 Ω, R2 = 500 Ω).

9 DC and AC parameters

This section summarizes the operating and measurement conditions, and the DC and AC characteristics.

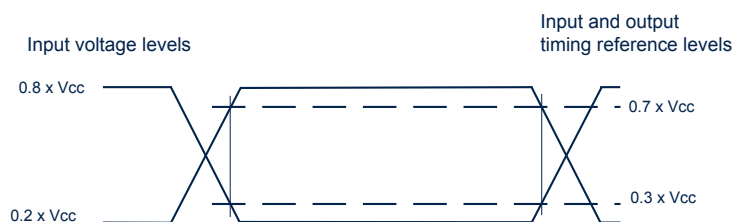
Table 13. Operating conditions

Symbol	Parameter	Min.	Max.	Unit
V_{CC}	Supply voltage	1.6	5.5	V
T_A	Ambient operating temperature	-40	85	°C
f_C	Operating clock frequency	-	1	MHz

Table 14. AC measurement conditions

Symbol	Parameter	Min.	Max.	Unit
C_{bus}	Load capacitance	-	100	pF
-	SCL input rise/fall time, SDA input fall time	-	50	ns
-	Input levels	0.2 V_{CC} to 0.8 V_{CC}		V
-	Input and output timing reference levels	0.3 V_{CC} to 0.7 V_{CC}		V

Figure 23. AC measurement I/O waveform



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Table 15. Input parameters

Symbol	Parameter	Test condition	Min.	Max.	Unit
$C_{IN}^{(1)}$	Input capacitance (SDA)	-	-	8	pF
$C_{IN}^{(1)}$	Input capacitance (other pins)	-	-	6	pF
$Z_L^{(2)}$	Input impedance ($\overline{WC}$) ⁽³⁾	$V_{IN} < 0.3 V_{CC}$	30	-	kΩ
$Z_H^{(2)}$		$V_{IN} > 0.7 V_{CC}$	500	-	kΩ

1. Specified by design – Not tested in production.
2. Evaluated by characterization – Not tested in production.
3. The memory is selected (after a start condition).

Table 16. Cycling performance by groups of four bytes

Symbol	Parameter	Test condition	Max.	Unit
Ncycle	Write cycle endurance ⁽¹⁾	$T_A \leq 25\text{ }^{\circ}\text{C}, V_{CC(\min)} < V_{CC} < V_{CC(\max)}$	4.000.000	Write cycle ⁽²⁾
		$T_A = 85\text{ }^{\circ}\text{C}, V_{CC(\min)} < V_{CC} < V_{CC(\max)}$	1.200.000	

1. The write cycle endurance is defined by characterization and qualification. For devices embedding the ECC functionality, the write cycle endurance is defined for a group of four bytes located at addresses $[4*N, 4*N+1, 4*N+2, 4*N+3]$ where N is an integer.
2. A Write cycle is executed when either a page write, a byte write, a write registers, a write identification page or a lock identification page instruction is decoded. When using the byte write, the page write or the write identification page, refer also to [Section 6.4: ECC \(error correction code\) and write cycling](#).

Table 17. Memory cell data retention

Parameter	Test condition	Min.	Unit
Data retention ⁽¹⁾	$T_A = 55\text{ }^{\circ}\text{C}$	200	Year

1. The data retention behaviour is checked in production, while the data retention limit is extracted from the characterization and qualification results.

Table 18. DC characteristics

Symbol	Parameter	Test conditions	Min.	Max.	Unit
I_{LI}	Input leakage current (SCL, SDA)	$V_{IN} = V_{SS}$ or V_{CC} device in standby mode	-	± 2	μA
I_{LO}	Output leakage current	SDA in high-Z, external voltage applied on SDA: V_{SS} or V_{CC}	-	± 2	μA
I_{CC}	Supply current (Read)	$f_C = 400 \text{ kHz}$	-	0.5 ⁽¹⁾	mA
		$f_C = 1 \text{ MHz}$	-	1 ⁽²⁾	mA
I_{CC0} ⁽³⁾	Supply current (write)	Averaged on T_W , $V_{CC} \leq 3.3 \text{ V}$	-	1 ⁽⁴⁾	mA
		Averaged on T_W , $V_{CC} > 3.3 \text{ V}$	-	1.5 ⁽⁵⁾	
I_{CC1}	Standby supply current	Device not selected ⁽⁶⁾ $V_{IN} = V_{SS}$ or V_{CC} ; $V_{CC} < 2.5 \text{ V}$	-	1 ⁽⁷⁾	μA
		Device not selected ⁽⁶⁾ $V_{IN} = V_{SS}$ or V_{CC} ; $V_{CC} \geq 2.5 \text{ V}$	-	2	μA
V_{IL}	Input low voltage (SCL, SDA, $\overline{WC}$)	$1.6 \text{ V} \leq V_{CC} < 2.5 \text{ V}$	-0.45	$0.25 V_{CC}$	V
		$2.5 \text{ V} \leq V_{CC} \leq 5.5 \text{ V}$	-0.45	$0.30 V_{CC}$	V
V_{IH}	Input high voltage (SCL, SDA)	$1.6 \text{ V} \leq V_{CC} < 2.5 \text{ V}$	$0.75 V_{CC}$	6.5	V
		$2.5 \text{ V} \leq V_{CC} \leq 5.5 \text{ V}$	$0.70 V_{CC}$	6.5	V
	Input high voltage ($\overline{WC}$)	$1.6 \text{ V} \leq V_{CC} < 2.5 \text{ V}$	$0.75 V_{CC}$	$V_{CC}+0.6$	V
		$2.5 \text{ V} \leq V_{CC} \leq 5.5 \text{ V}$	$0.70 V_{CC}$	$V_{CC}+0.6$	V
V_{OL}	Output low	$I_{OL} = 1 \text{ mA}$, $V_{CC} = 1.6 \text{ V}$	-	0.2	V
		$I_{OL} = 2.1 \text{ mA}$, $V_{CC} = 2.5 \text{ V}$ or	-	0.4	V
		$I_{OL} = 3 \text{ mA}$, $V_{CC} = 5.5 \text{ V}$	-	0.4	V

1. Typical value at 1.8 V (evaluated by characterization, not tested in production) is 70 μA .
2. Typical value at 1.8 V (evaluated by characterization, not tested in production) is 100 μA .
3. Evaluated by characterization - Not tested in production.
4. Typical value at 1.8 V (evaluated by characterization, not tested in production) is 360 μA .
5. Typical value at 3.3 V (evaluated by characterization, not tested in production) is 650 μA .
6. The device is not selected after power-up, after a read instruction (after the stop condition), or after the completion of the internal write cycle t_W (t_W is triggered by the correct decoding of a write instruction).
7. Typical value at 1.8 V and 25 °C (evaluated by characterization, not tested in production) is 330 nA.

Table 19. AC characteristics in Fast-mode

Symbol	Alt.	Parameter	Min.	Max.	Unit
f_C	f_{SCL}	Clock frequency	-	400	kHz
t_{CHCL}	t_{HIGH}	Clock pulse width high	600	-	ns
t_{CLCH}	t_{LOW}	Clock pulse width low	1300	-	ns
$t_{QL1QL2}^{(1)}$	t_F	SDA (out) fall time	20 ⁽²⁾	300	ns
$t_{XH1XH2}^{(1)}$	t_R	Input signal rise time	(3)	(3)	ns
$t_{XL1XL2}^{(1)}$	t_F	Input signal fall time	(3)	(3)	ns
t_{DXCH}	$t_{SU:DAT}$	Data in setup time	100	-	ns
t_{CLDX}	$t_{HD:DAT}$	Data in hold time	0	-	ns
$t_{CLQX}^{(4)}$	t_{DH}	Data out hold time	100	-	ns
$t_{CLQV}^{(5)}$	t_{AA}	Clock low to next data valid (access time)	-	900	ns
t_{CHDL}	$t_{SU:STA}$	Start condition setup time	600	-	ns
t_{DLCL}	$t_{HD:STA}$	Start condition hold time	600	-	ns
t_{CHDH}	$t_{SU:STO}$	Stop condition set up time	600	-	ns
t_{DHDL}	t_{BUF}	Time between Stop condition and next Start condition	1300	-	ns
$t_{WLDL}^{(1)(6)}$	$t_{SU:WC}$	$\overline{WC}$ set up time (before the start condition)	0	-	μs
$t_{DWHH}^{(1)(7)}$	$t_{HD:WC}$	$\overline{WC}$ hold time (after the stop condition)	1	-	μs
t_W	t_{WR}	Write cycle time	-	4 ⁽⁸⁾	ms
$t_{NS}^{(1)}$	-	Pulse width ignored (input filter on SCL and SDA) - single glitch	-	50	ns
$t_{WU}^{(9)(10)}$	-	Wake up time	-	5	μs

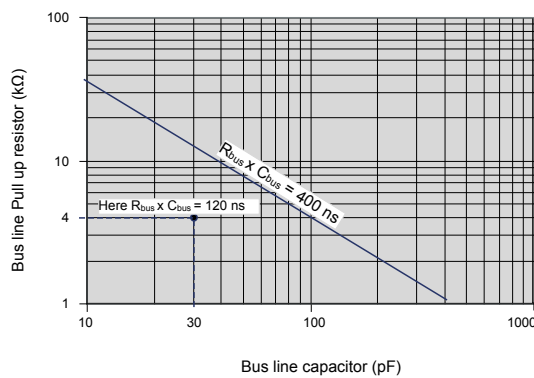
1. Evaluated by characterization - Not tested in production.
2. With $C_L = 10$ pF.
3. There is no minimum or maximum values for the input signal rise and fall times. It is however recommended by the I²C specification that the input signal rise and fall times be more than 20 ns and less than 300 ns when $f_C < 400$ kHz.
4. To avoid spurious start and stop conditions, a minimum delay is placed between SCL = 1 and the falling or rising edge of SDA.
5. t_{CLQV} is the time (from the falling edge of SCL) required by the SDA bus line to reach either 0.3V_{CC} or 0.7V_{CC}, assuming that the $R_{bus} \times C_{bus}$ time constant is within the values specified in Figure 24.
6. $\overline{WC} = 0$ setup time condition to enable the execution of a write command.
7. $\overline{WC} = 0$ hold time condition to enable the execution of a write command.
8. 3.1 ms typical.
9. Specified by design - Not tested in production.
10. Wake up time: Delay between the V_{CC(min)} stable and the first accepted command.

Table 20. AC characteristics in Fast-mode Plus

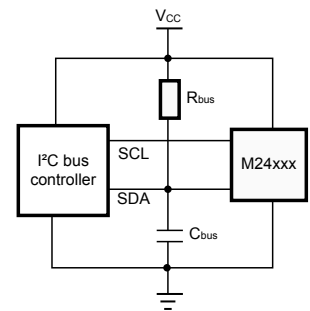
Symbol	Alt.	Parameter	Min.	Max.	Unit
f_C	f_{SCL}	Clock frequency	-	1	MHz
t_{CHCL}	t_{HIGH}	Clock pulse width high	260	-	ns
t_{CLCH}	t_{LOW}	Clock pulse width low	500	-	ns
t_{XH1XH2}	t_R	Input signal rise time	(1)	(1)	ns
t_{XL1XL2}	t_F	Input signal fall time	(1)	(1)	ns
$t_{QL1QL2}^{(2)}$	t_F	SDA (out) fall time	20 ⁽³⁾	120	ns
t_{DXCH}	$t_{SU:DAT}$	Data in setup time	50	-	ns
t_{CLDX}	$t_{HD:DAT}$	Data in hold time	0	-	ns
$t_{CLQX}^{(4)}$	t_{DH}	Data out hold time	100	-	ns
$t_{CLQV}^{(5)}$	t_{AA}	Clock low to next data valid (access time)	-	450	ns
t_{CHDL}	$t_{SU:STA}$	Start condition setup time	250	-	ns
t_{DLCL}	$t_{HD:STA}$	Start condition hold time	250	-	ns
t_{CHDH}	$t_{SU:STO}$	Stop condition setup time	250	-	ns
t_{DHDL}	t_{BUF}	Time between Stop condition and next Start condition	500	-	ns
$t_{WLDL}^{(2)(6)}$	$t_{SU:WC}$	$\overline{WC}$ set up time (before the start condition)	0	-	μs
$t_{DHWL}^{(2)(7)}$	$t_{HD:WC}$	$\overline{WC}$ hold time (after the stop condition)	1	-	μs
t_W	t_{WR}	Write cycle time	-	4 ⁽⁸⁾	ms
$t_{NS}^{(2)}$	-	Pulse width ignored (input filter on SCL and SDA)	-	50	ns
$t_{WU}^{(9)(10)}$	-	Wake up time	-	5	μs

1. There is no minimum or maximum values for the input signal rise and fall times. It is however recommended by the I²C specification that the input signal rise and fall times be less than 120 ns when $f_C < 1$ MHz.
2. Evaluated by characterization - Not tested in production.
3. With $CL = 10$ pF.
4. To avoid spurious Start and Stop conditions, a minimum delay is placed between $SCL = 1$ and the falling or rising edge of SDA.
5. t_{CLQV} is the time (from the falling edge of SCL) required by the SDA bus line to reach either $0.3 V_{CC}$ or $0.7 V_{CC}$, assuming that the $R_{bus} \times C_{bus}$ time constant is within the values specified in Figure 25.
6. $\overline{WC} = 0$ setup time condition to enable the execution of a write command.
7. $\overline{WC} = 0$ hold time condition to enable the execution of a write command.
8. 3.1 ms typical.
9. Specified by design - Not tested in production.
10. Wake up time: Delay between the $V_{CC(min)}$ stable and the first accepted commands.

Figure 24. R_{bus} value versus bus parasitic capacitance (C_{bus}) for an I^2C_{bus} ($f_c = 400$ kHz)

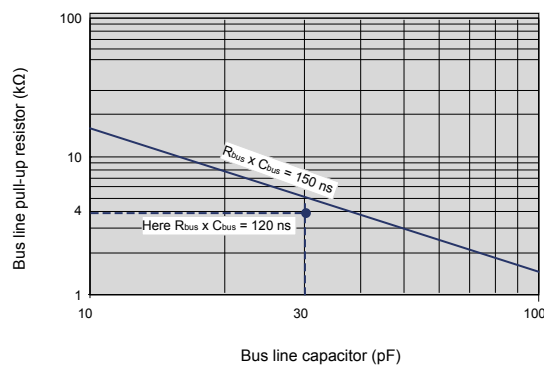


The $R_{bus} \times C_{bus}$ time constant must be below the 400 ns time constant line displayed on the left

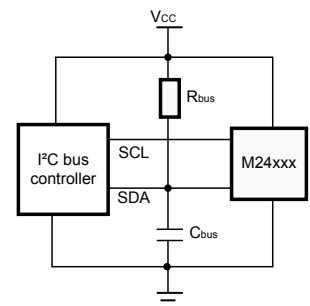


DT137916V5

Figure 25. R_{bus} value versus bus parasitic capacitance (C_{bus}) for an I^2C bus ($f_c = 1$ MHz)



The $R_{bus} \times C_{bus}$ time constant must be below the 150 ns time constant line displayed on the left



DT19745V8

Figure 26. AC waveforms



10 Package information

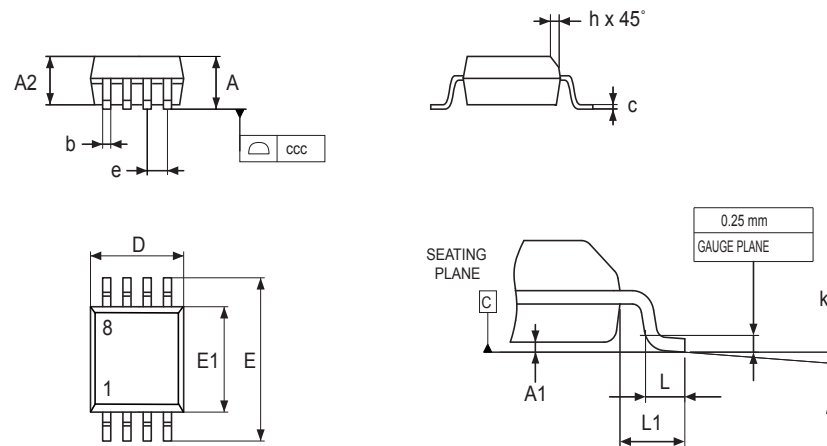
To meet environmental requirements, ST offers these devices in different grades of ECOPACK packages, depending on their level of environmental compliance. ECOPACK specifications, grade definitions, and product status are available at: www.st.com. ECOPACK is an ST trademark.

For die information concerning the M24512E delivered in unsawn wafer, contact your nearest ST Sales Office.

10.1 SO8N package information

This SO8N is an 8-lead, 4.9 x 6 mm, plastic small outline, 150 mil body width package.

Figure 27. SO8N - Outline



1. Drawing is not to scale.

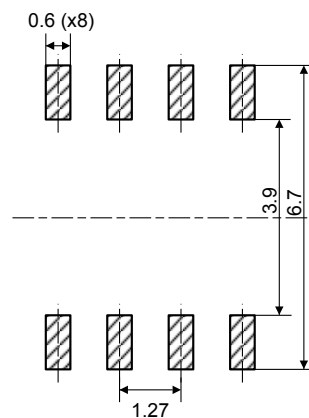
O7_SO8_ME_V2

Table 21. SO8N - Mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min.	Typ.	Max.	Min.	Typ.	Max.
A	-	-	1.750	-	-	0.0689
A1	0.100	-	0.250	0.0039	-	0.0098
A2	1.250	-	-	0.0492	-	-
b	0.280	-	0.480	0.0110	-	0.0189
c	0.170	-	0.230	0.0067	-	0.0091
D ⁽²⁾	4.800	4.900	5.000	0.1890	0.1929	0.1969
E	5.800	6.000	6.200	0.2283	0.2362	0.2441
E1 ⁽³⁾	3.800	3.900	4.000	0.1496	0.1535	0.1575
e	-	1.270	-	-	0.0500	-
h	0.250	-	0.500	0.0098	-	0.0197
k	0°	-	8°	0°	-	8°
L	0.400	-	1.270	0.0157	-	0.0500
L1	-	1.040	-	-	0.0409	-
ccc	-	-	0.100	-	-	0.0039

1. Values in inches are converted from mm and rounded to four decimal digits.
2. Dimension D does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
3. Dimension E1 does not include interlead flash or protrusions. Interlead flash or protrusions shall not exceed 0.25 mm per side.

Note: The package top may be smaller than the package bottom. Dimensions D and E1 are determined at the outermost extremes of the plastic body exclusive of mold flash, tie bar burrs, gate burrs, and interleads flash, but including any mismatch between the top and bottom of the plastic body. The measurement side for mold flash, protrusions, or gate burrs is the bottom side.

Figure 28. SO8N - Footprint example


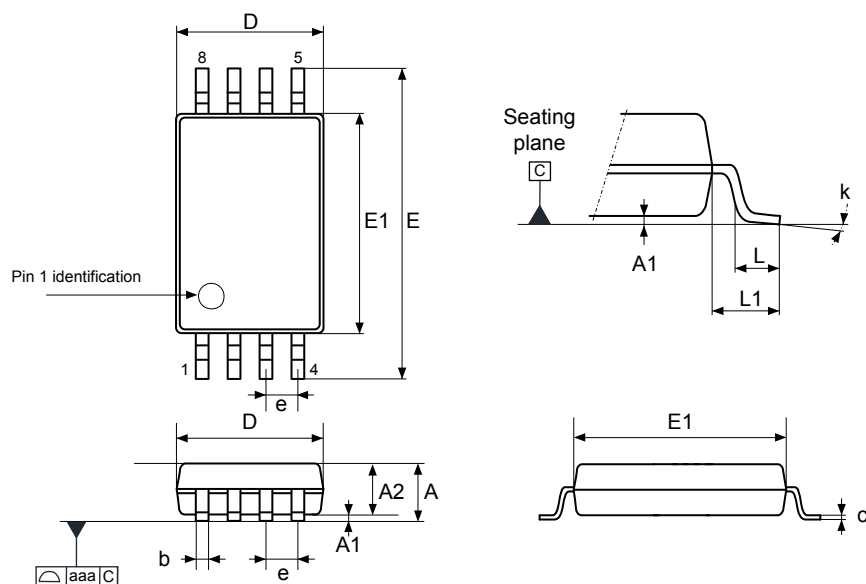
07_SO8N_FP_V2

1. Dimensions are expressed in millimeters.

10.2 TSSOP8 package information

This TSSOP is an 8-lead, 3 x 6.4 mm, 0.65 mm pitch, thin shrink small outline package.

Figure 29. TSSOP8 – Outline



DT_6P_A_TSSOP8_ME_V4

1. Drawing is not to scale.

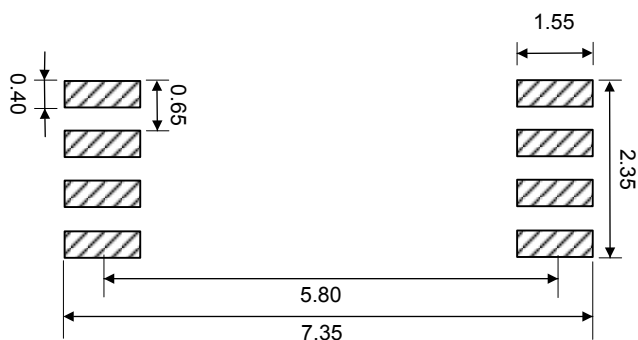
Table 22. TSSOP8 - Mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min.	Typ.	Max.	Min.	Typ.	Max.
A	-	-	1.200	-	-	0.0472
A1	0.050	-	0.150	0.0020	-	0.0059
A2	0.800	1.000	1.050	0.0315	0.0394	0.0413
b	0.190	-	0.300	0.0075	-	0.0118
c	0.090	-	0.200	0.0035	-	0.0079
D ⁽²⁾	2.900	3.000	3.100	0.1142	0.1181	0.1220
e	-	0.650	-	-	0.0256	-
E	6.200	6.400	6.600	0.2441	0.2520	0.2598
E1 ⁽³⁾	4.300	4.400	4.500	0.1693	0.1732	0.1772
L	0.450	0.600	0.750	0.0177	0.0236	0.0295
L1	-	1.000	-	-	0.0394	-
k	0°	-	8°	0°	-	8°
aaa	-	-	0.100	-	-	0.0039

1. Values in inches are converted from mm and rounded to four decimal digits.
2. Dimension D does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
3. Dimension E1 does not include interlead flash or protrusions. Interlead flash or protrusions shall not exceed 0.25 mm per side.

Note: The package top may be smaller than the package bottom. Dimensions D and E1 are determined at the outermost extremes of the plastic body exclusive of the mold flash, tie bar burrs, gate burrs, and interleads flash, but including any mismatch between the top and bottom of the plastic body. The measurement side for the mold flash, protrusions, or gate burrs is the bottom side.

Figure 30. TSSOP8 – Footprint example



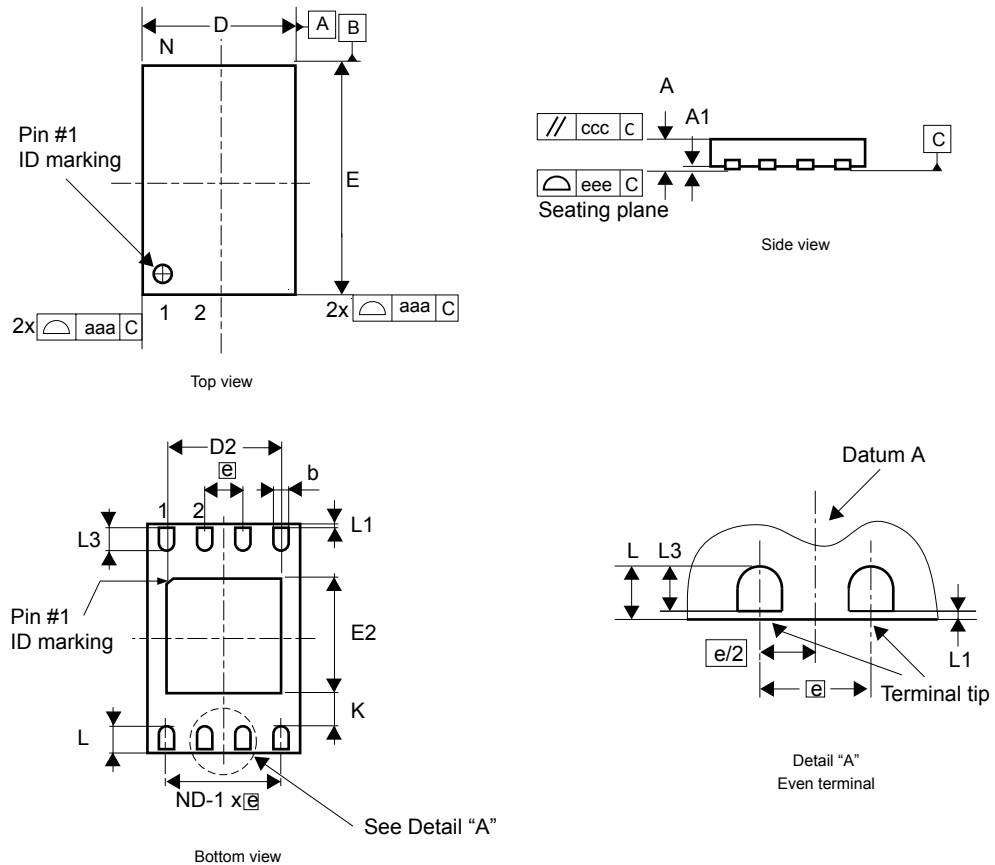
DT_6P_TSSOP8_FP_V2

1. Dimensions are expressed in millimeters.

10.3 UFDFPN8 (DFN8) package information

This UFDFPN is an 8-lead, 2 x 3 mm, 0.5 mm pitch ultrathin profile fine pitch dual flat package.

Figure 31. UFDFPN8 - Outline



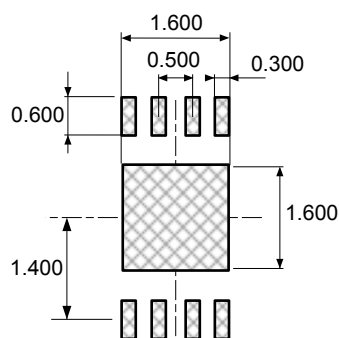
1. The maximum package warpage is 0.05 mm.
2. Exposed copper is not systematic and can appear partially or totally according to the cross section.
3. Drawing is not to scale.
4. The central pad (the area E2 by D2 in the above illustration) must be either connected to V_{SS} or left floating (not connected) in the end application.

ZWb_UFDFPN8_ME_V2

Table 23. UFDFPN8 - Mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	0.450	0.550	0.600	0.0177	0.0217	0.0236
A1	0.000	0.020	0.050	0.0000	0.0008	0.0020
b ⁽²⁾	0.200	0.250	0.300	0.0079	0.0098	0.0118
D	1.900	2.000	2.100	0.0748	0.0787	0.0827
D2	1.200	-	1.600	0.0472	-	0.0630
E	2.900	3.000	3.100	0.1142	0.1181	0.1220
E2	1.200	-	1.600	0.0472	-	0.0630
e	-	0.500	-	-	0.0197	-
K	0.300	-	-	0.0118	-	-
L	0.300	-	0.500	0.0118	-	0.0197
L1	-	-	0.150	-	-	0.0059
L3	0.300	-	-	0.0118	-	-
aaa	-	-	0.150	-	-	0.0059
bbb	-	-	0.100	-	-	0.0039
ccc	-	-	0.100	-	-	0.0039
ddd	-	-	0.050	-	-	0.0020
eee ⁽³⁾	-	-	0.080	-	-	0.0031

1. Values in inches are converted from mm and rounded to four decimal digits.
2. Dimension b applies to the plated terminal and is measured between 0.15 and 0.30 mm from the terminal tip.
3. Applied for exposed die paddle and terminals. Exclude embedding part of the exposed die paddle from measuring.

Figure 32. UFDFPN8 - Footprint example


1. Dimensions are expressed in millimeters.

ZWb_UFDFPN8_FP_V2

11 Ordering information

Table 24. Ordering information scheme

Example:	M24	512E	-	F	MN	6	T	P
Device type								
M24 = I ² C serial access EEPROM								
Device function								
512E = 512-Kbit (64 K x 8 bit)								
Operating voltage⁽¹⁾								
F = V _{CC} = 1.6 V to 5.5 V								
Package⁽²⁾								
MN = SO8 (150 mil width)								
DW = TSSOP8 (169 mil width)								
MC = UDFPN8 (DFN8)								
Device grade								
6 = Industrial device tested with standard test flow over -40 to 85 °C								
Option								
T = Tape and reel packing								
Blank = tube packing								
Plating technology and device address								
P or G = ECOPACK2 and device address 000								
1 to 7 = ECOPACK2 and preprogrammed device address ⁽³⁾								

1. Products with finished good M24512EFMN6TPV0B or with top marking 24512FP and KxxxV on the second line are associated to the datasheet M24512E-F revision 1 - October 07, 2022.
2. ECOPACK2 (RoHS compliant and free of brominated, chlorinated, and antimony oxide flame retardants).
3. See Table 6. Preprogrammed device address.

Table 25. Ordering information scheme (unsawn wafer)

Example:	M24	512E	- F	V	W	20	I	/90
Device type								
M24 = I ² C serial access EEPROM								
Device function								
512E = 512-Kbit (64 K x 8 bit)								
Operating voltage								
F = V _{CC} = 1.6 V to 5.5 V								
Process								
V = Manufacturing technology code								
Delivery form								
W = Unsawn wafer								
Wafer thickness								
20 = Non backlapped wafer								
Wafer testing								
I = Inkless test								
Grade								
/90 = -40 to 85 °C								

Note: For a list of available options (memory, package, and so on) or for further information on any aspect of this device, contact your nearest ST sales office.

Note: Parts marked as “ES” or “E” are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST’s Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

Revision history

Table 26. Document revision history

Date	Revision	Changes
07-Oct-2022	1	Initial release
13-Feb-2023	2	Added TSSOP8 and UFDFPN8 packages. Note 1 in Section 11: Ordering information Updated: • minimum Voltage supply from 2.5 V to 1.6 V • Section 4.1: Device type identifier register (DTI), Section 4.2: Configurable device address register (CDA), Section 4.3: Software write protection register (SWP), Section 6.2.3: Write operation on SWP register • Table 15. Input parameters, Table 18. DC characteristics
26-Oct-2023	3	Updated: • Features • Section 1: Description • Section 4.2: Configurable device address register (CDA) • Section 6.2.2: Write operation on CDA register • Table 18. DC characteristics • Section 11: Ordering information Added: • Section 4.2: Configurable device address register (CDA)
30-Aug-2024	4	Updated: • Features • Section 1: Description • Section 4.2: Configurable device address register (CDA) • Section 4.2: Configurable device address register (CDA) • Section 4.3: Software write protection register (SWP) • Section 7: Initial delivery state • Section 10: Package information Added: • Section 11: Ordering information
02-Jun-2026	5	Updated: • Section Cover image • Section 1: Description • Section 2.3: Write control ($\overline{WC}$) • Section 2.5.3: Device reset • Section 4.1: Device type identifier register (DTI) • Section 4.2: Configurable device address register (CDA) • Section 4.3: Software write protection register (SWP) • Section 6.1.1: Byte write • Section 6.2.4: Write operation on identification page

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